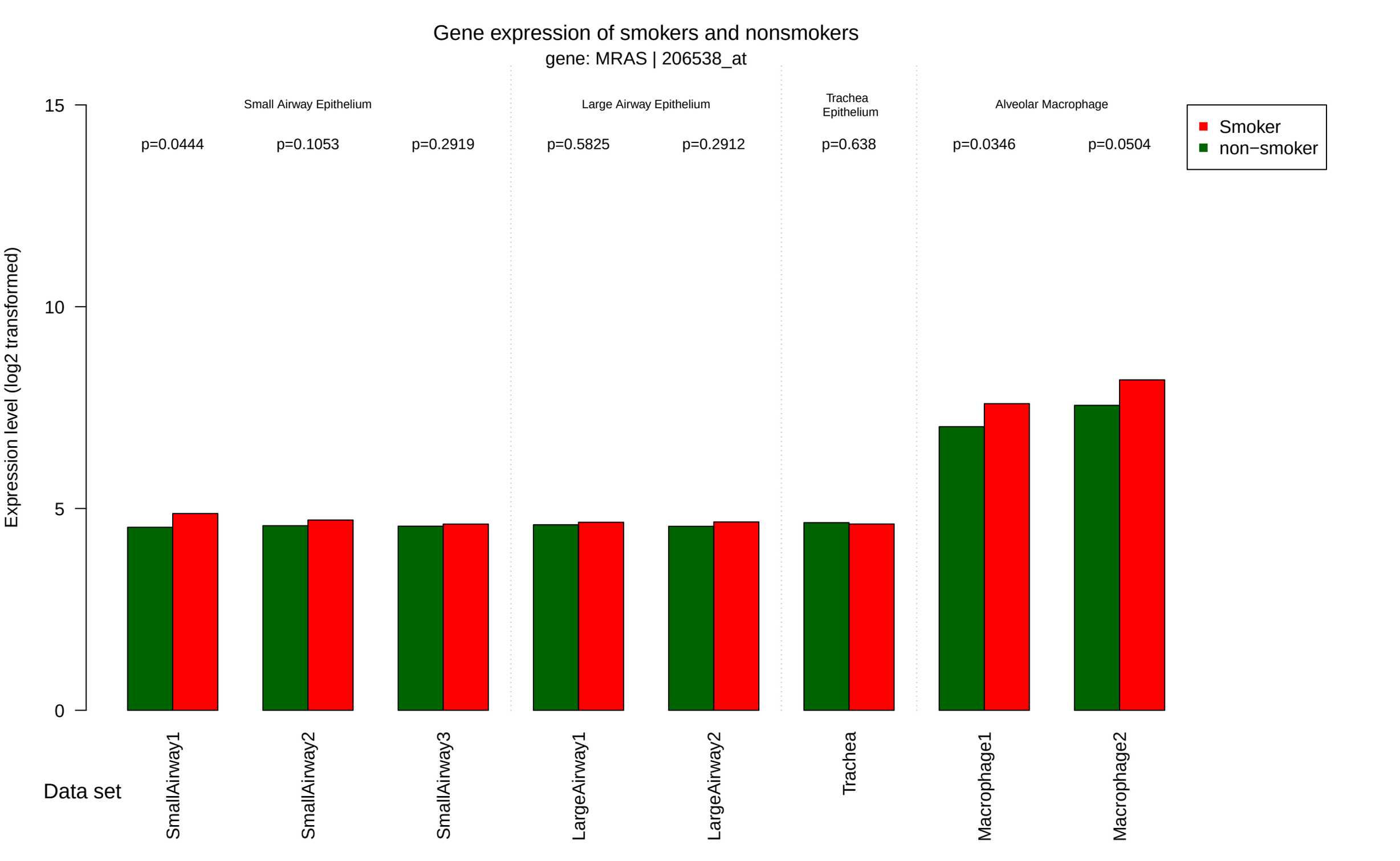
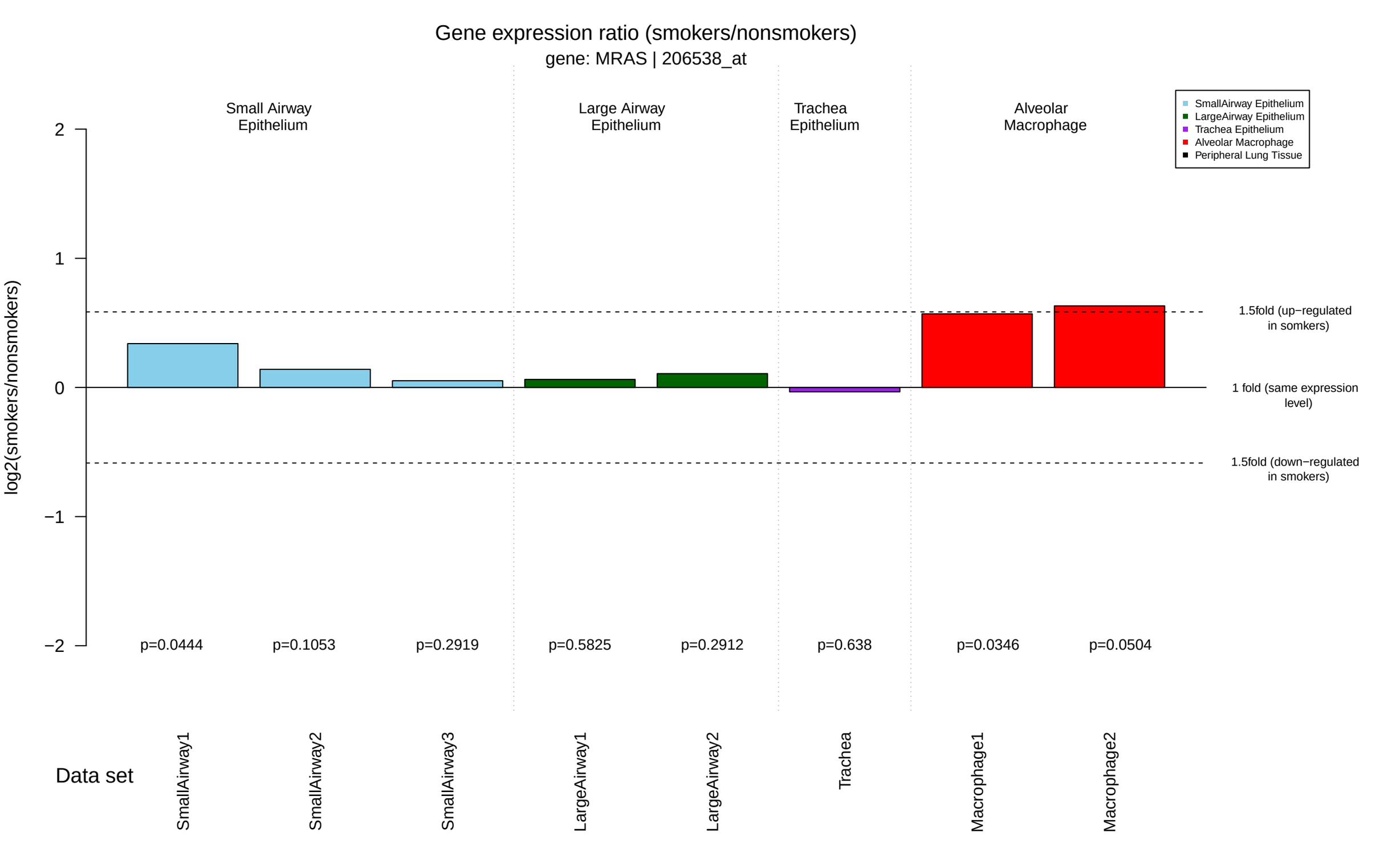
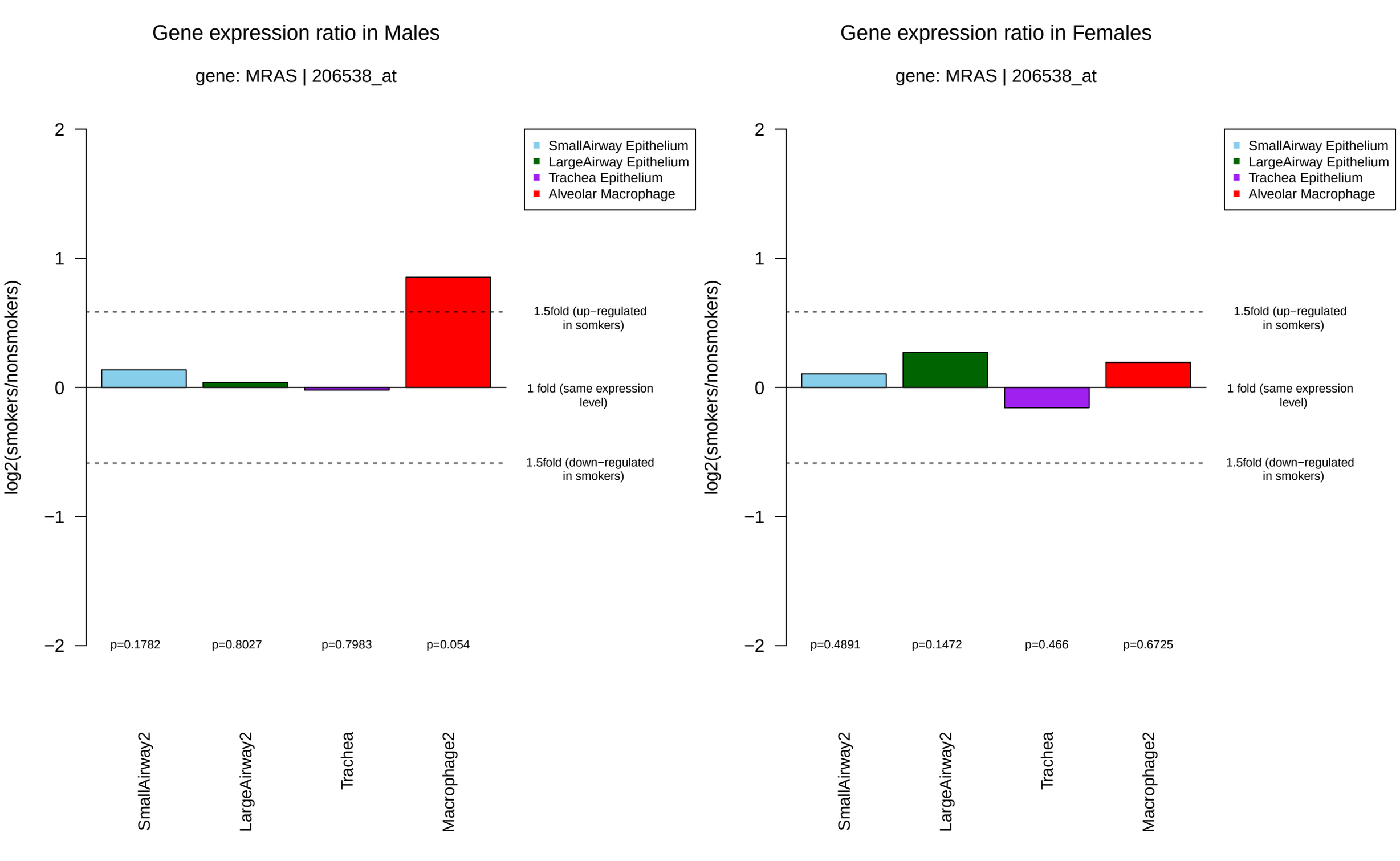
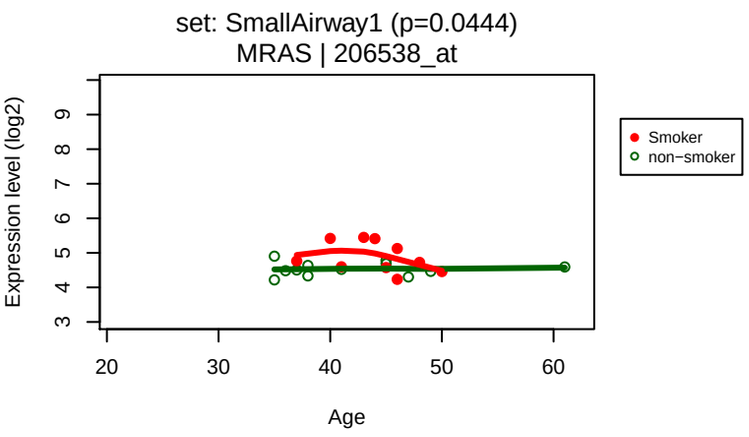
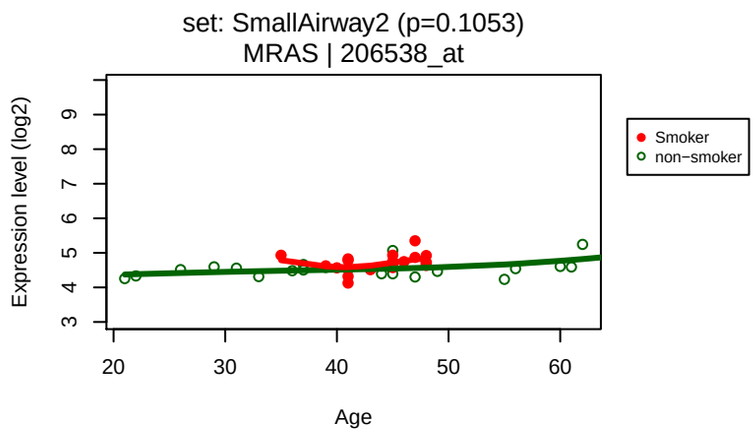
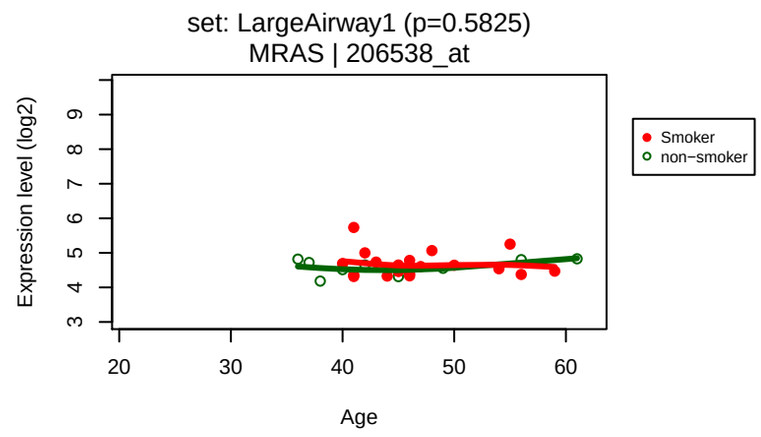

Gene expression of smokers and nonsmokers
gene: MRAS | 206538_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0444
p=0.1053
p=0.2919
p=0.5825
p=0.2912
p=0.638
p=0.0346
p=0.0504
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

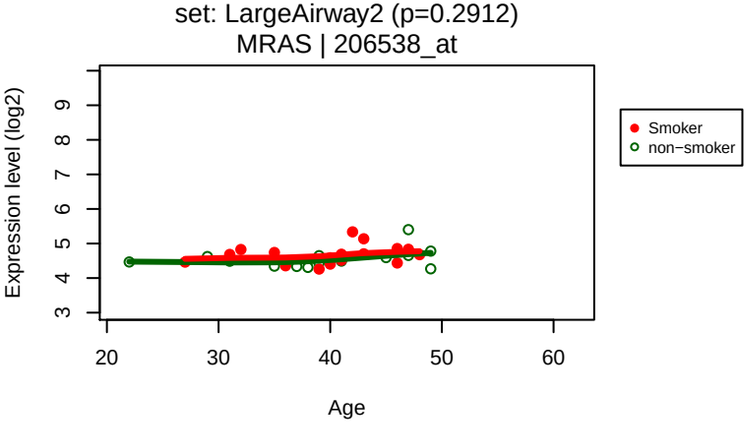

Gene expression ratio (smokers/nonsmokers)
gene: MRAS | 206538_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0444
p=0.1053
p=0.2919
p=0.5825
p=0.2912
p=0.638
p=0.0346
p=0.0504
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

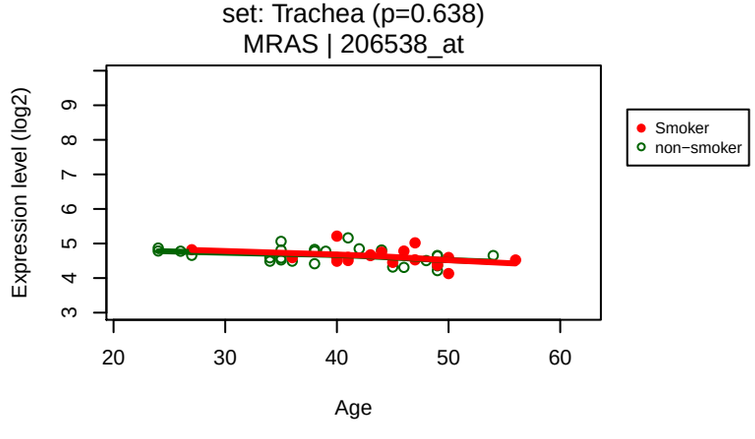

Gene expression ratio in Males
gene: MRAS | 206538_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1782
p=0.8027
p=0.7983
p=0.054
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRAS | 206538_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4891
p=0.1472
p=0.466
p=0.6725
SmallAirway2
LargeAirway2
Trachea
Macrophage2

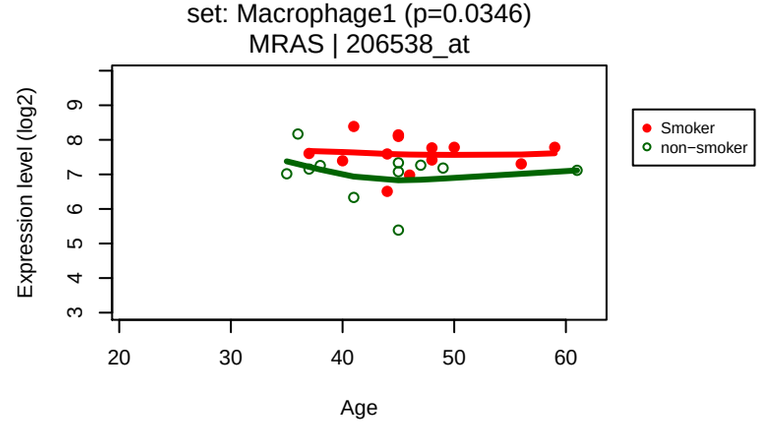

set: SmallAirway1 (p=0.0444)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker

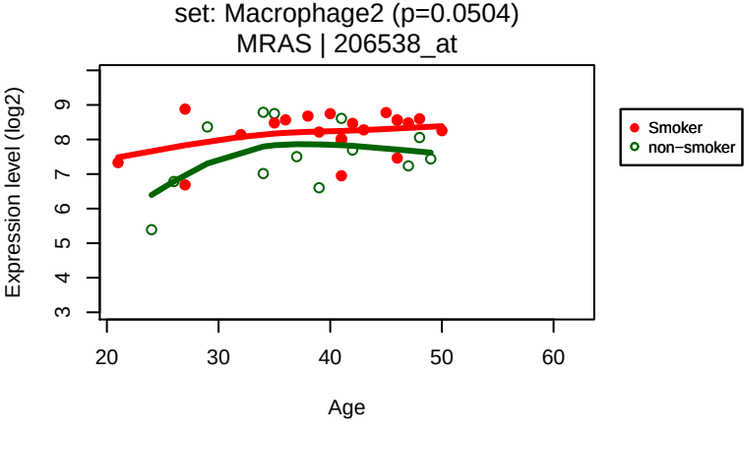

set: SmallAirway2 (p=0.1053)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker

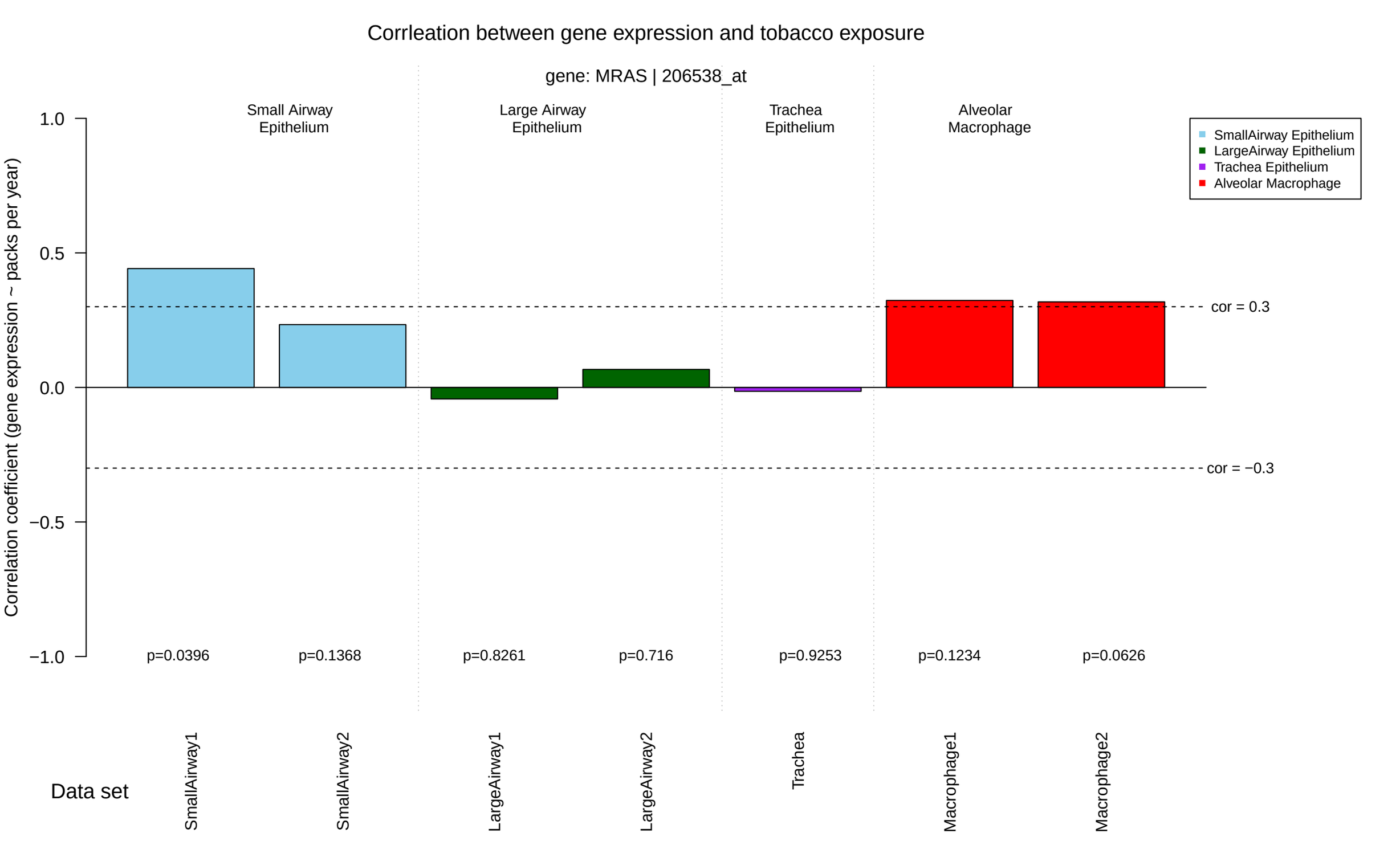

set: LargeAirway1 (p=0.5825)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2912)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.638)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0346)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0504)
MRAS | 206538_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRAS | 206538_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0396
p=0.1368
p=0.8261
p=0.716
p=0.9253
p=0.1234
p=0.0626
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2